

Budapest University of Technology and Economics
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M2 - Basic circuits for operational amplifiers

Lab report

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Abstract

Basic properties of operational amplifier circuits are investigated. Gain and saturation of the inverting amplifier is determined. A summing amplifier is built and its performance is investigated. Behaviour of voltage follower is checked against sinusoidal signals and its frequency-dependence is investigated. Performance of differentiator and integrator circuits are examined. Our results are in accordance with the theory between ranges of error.

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1 Introduction

An operational amplifier (op-amp) is a device, having two inputs (inverting and non-inverting) and an output, and it amplifies the difference of voltages applied on its inputs, say if the voltage on the non-inverting input is U_1 and the voltage on the inverting input is U_2, then the output voltage is

$$U_{\text{out}} = A\,(U_1 - U_2)\,, \tag{1}$$

where A is the gain of the amplifier. For an ideal op-amp the gain is ∞, while for a realistic one it is still very high ($\sim 10^6$). The op-amp is driven by external supply voltages $U_\pm$, typically in a symmetric layout ($\pm 5\,\text{V}$, $\pm 9\,\text{V}$ or $\pm 12\,\text{V}$, etc). The maximum value of the output voltage for an ideal device is exactly the supply value, its reach is called the *saturation*. Between the saturation levels we can find the *linear* range by (1). Then the op-amp alone is basically a comparator, giving the negative supply if U_2 and the positive supply if U_1 is bigger.
The input resistance of the amplifier is very high (∞ for the ideal and MΩ for realistic) and its output resistance is very low (0 for ideal and few Ω-s for realistic). Other important properties are the bandwidth and the slew rate (rate of change in voltage by time).
In realistic situations a feedback is used between the input and output of the amplifier. The type and value of the feedback determines the type and gain of the circuit.

1.1 Inverting amplifier

The inverting amplifier is shown on Figure 1.

Using the fact that no current flows into the amplifier, then $I = U_{\text{in}}/R_{\text{s}} = -U_{\text{out}}/R_{\text{v}}$, then

$$A_{\text{id}} = \frac{U_{\text{out}}}{U_{\text{in}}} = -\frac{R_{\text{v}}}{R_{\text{s}}} \tag{2}$$

for an ideal amplifier.

1.2 Summing amplifier

The summing amplifier is shown on Figure 2. Here

$$-\frac{U_1}{R_1} - \frac{U_2}{R_2} = \frac{U_{\text{out}}}{R_{\text{v}}}, \tag{3}$$

then

$$U_{\text{out}} = -\left(\frac{R_{\text{v}}}{R_1}U_1 + \frac{R_{\text{v}}}{R_2}U_2\right), \tag{4}$$

say it provides the weighted sum of input voltages.

1.3 Voltage follower

The voltage follower is shown on Figure 3.

The circuit is required to keep the amplitude and the phase of the input. This circuit is used as a buffer for measuring small signals caused by the high input resistance of the amplifier.

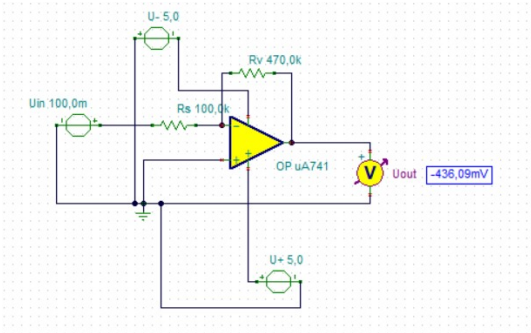


Figure 1: Diagram of inverting amplifier circuit. Here Rv is the feedback resistor. U+ and U- provide the supply voltages and Uin is the input voltage. The type of the amplifier is a µA741, and the input voltage is 100 mV, then the output can be read from the Uout device being $-436.09\,\mathrm{mV}$ with $R_\mathrm{v} = 470\,\mathrm{k\Omega}$ and $R_\mathrm{s} = 100\,\mathrm{k\Omega}$ and $U_\pm = \pm 5\,\mathrm{V}$, say the amplification is about $A = -4.4$, while the ideal value gives $A_\mathrm{id} = -4.7$.

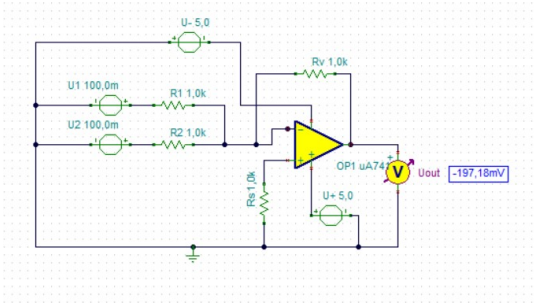


Figure 2: Diagram of summing amplifier circuit. Here Rv is the feedback resistor, R1 and R2 are the input resistances. U+ and U- provide the supply voltages, while U1 and U2 are the input voltages. The type of the amplifier is a µA741, and both input voltages are 100 mV, then the output can be read from the Uout device being $-197.18\,\mathrm{mV}$ with $R_\mathrm{v} = R_\mathrm{s} = R_1 = R_2 = 1\,\mathrm{k\Omega}$ and $U_\pm = \pm 5\,\mathrm{V}$, say the circuit really sums them up with an inversion, while $-200\,\mathrm{mV}$ is expected.

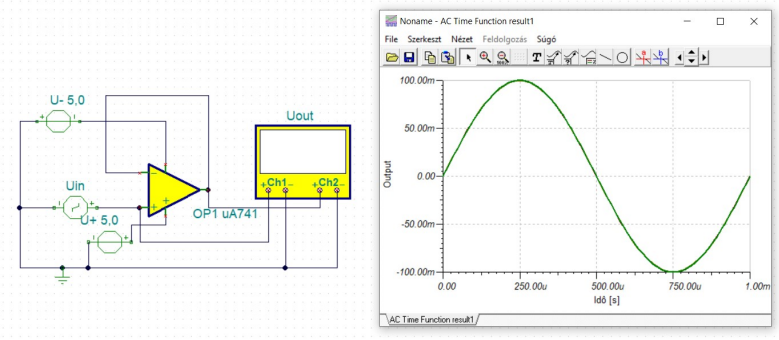


Figure 3: Diagram of voltage follower. U+ and U- provide the supply voltages and Uin is the input voltage, being a 100 mV and 1 kHz sinusoidal signal. The type of the amplifier is a µA741, and the input voltage $U_\pm = \pm 5\,\mathrm{V}$. The time-diagram is on the right, showing the following property of the circuit, say the output (green) is the same as the input (red).

1.4 Differentiator

The differentiator is shown on Figure 4.
Here the current flowing through the capacitor is

$$I = C\frac{\mathrm{d}U_\mathrm{in}}{\mathrm{d}t} = -\frac{U_\mathrm{out}}{R_\mathrm{v}}, \tag{5}$$

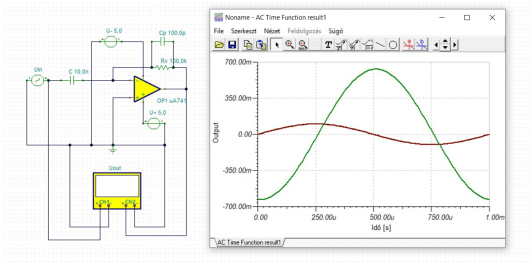


Figure 4: Diagram of differentiator circuit. Here Rv is the feedback resistor and Cp is the feedback capacitor, while C is the input capacitor. U+ and U- provides the supply voltages and Uin is the input voltage, being a 100 mV and 1 kHz sinusoidal signal. The type of the amplifier is a µA741 with $U_{\pm} = \pm 5\,\mathrm{V}$. Here $R_\mathrm{v} = 100\,\mathrm{k\Omega}$, $C_\mathrm{p} = 100\,\mathrm{pF}$ and $C = 10\,\mathrm{nF}$. the time-diagram is on the right. Here we can see a negative cosine function (green), which is the derivative of the sine (red) with an inversion, with an expected amplitude ratio of 2π.

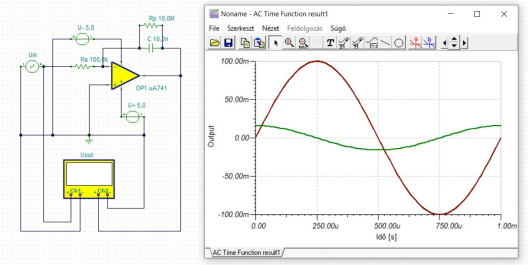


Figure 5: Diagram of integrator circuit. Here Rp is the feedback resistor and C is the feedback capacitor, while Rs is the input resistor. U+ and U- provides the supply voltages and Uin is the input voltage, being a 100 mV and 1 kHz sinusoidal signal. The type of the amplifier is a µA741 with $U_{\pm} = \pm 5\,\mathrm{V}$. Here $R_\mathrm{p} = 10\,\mathrm{M\Omega}$, $C = 10\,\mathrm{nF}$ and $R_\mathrm{s} = 100\,\mathrm{k\Omega}$. the time-diagram is on the right. Here we can see a cosine function (green), which is the integral of the sine (red) with an inversion and an expected amplitude ratio of $(2\pi)^{-1}$.

say the output is proportional to the derivative of the input. The amplification for a sinusoidal signal is $2\pi f C R_\mathrm{v}$, where f is the frequency.

1.5 Integrator

The integrator is shown on Figure 5. Here

$$C\frac{\mathrm{d}U_\mathrm{out}}{\mathrm{d}t} = -\frac{U_\mathrm{in}}{R_\mathrm{s}} \qquad \Rightarrow \qquad U_\mathrm{out} = -\frac{1}{R_\mathrm{s}C}\int_0^t U_\mathrm{in}(t)\,\mathrm{d}t. \tag{6}$$

1.6 Miller-effect

Since there are always stray capacities inside the circuit there is always a capacitive feedback between the input and the output of the amplifier. If it has a value of C and $U_\mathrm{out} = AU_\mathrm{in}$, the current flowing through the stray capacitance is

$$I_\mathrm{c} = -C\frac{\mathrm{d}}{\mathrm{d}t}\left(-AU_\mathrm{in} - U_\mathrm{in}\right) = -C(1+A)\frac{\mathrm{d}U_\mathrm{in}}{\mathrm{d}t}, \tag{7}$$

say the stray capacitance behaves as a $C(1+A)$ capacity connected to the input, which can have a very high value, since A is big. This is the Miller-effect, which causes the decrease of gain at high frequencies (the derivative is proportional to the frequency).

2 List of applied instruments

- Cables
- TL071 operational amplifier [2]
- Hameg HM 8040 triple power supply [3]
- Hameg HM 8011 digital multimeter [4]
- Hameg HM 8030 function generator [5]
- Siglent SDS 1072 CML oscilloscope [6]
- Breadboard
- Resistors, capacitors

3 Measurement exercises

3.1 M1 - Inverting amplifier

In this exercise we have built the inverting amplifier circuit on the breadboard shown on Figure 1 and introduced in Section 1.1. Here the amplifier was the TL071, $R_\mathrm{v} = 680\,\mathrm{k\Omega}$, $R_\mathrm{s} = 100\,\mathrm{k\Omega}$.
At first, we applied $U_\mathrm{in} = 100\,\mathrm{mV}$ to the input, and increased it in 100 mV steps. The theoretical gain of the amplifier can be calculated using (2), here it is $A = -R_\mathrm{v}/R_\mathrm{s} = -6.8$. The measured output-input voltages are shown on Figure 6.

The linear part of the characteristics is expected being in the range $U_\mathrm{in} \in [0; 1\,\mathrm{V}]$, since the supply voltage is $\pm 5\,\mathrm{V}$. Consequently, we have fitted a line on that range, shown on Figure 6. The measured gain can be read out from the slope of the line, giving $A = -6.19 \pm 0.131$, where this error comes from the linear regression on the measured data points. The real error is much bigger, since the error of the input is $\pm 50\,\mathrm{mV}$, then by Gaussian error propagation the averaged error of the amplification is $\Delta A = \pm 1.2$, say the measured amplification of the system is $A = -6.2 \pm 1.2$, which is in accordance with the calculated value.

3.2 M2 - Summing amplifier

In this exercise we built the summing amplifier shown on Figure 2. We have used $R_\mathrm{v} = R_\mathrm{s} = 1\,\mathrm{k\Omega}$ and $R_1 = R_2 = 680\,\Omega$. We have applied different U_1 and U_2 values and measured the input of the system. Our results for the output voltage is shown in Table 1. We can see, that circuit really provides the weighted sum of the two inputs, since the calculated and measured values are match between ranges of error.

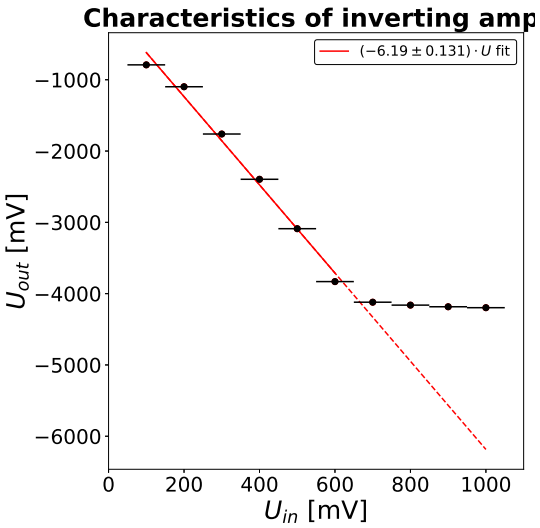


Figure 6: Output-input voltages on the inverting amplifier. The gain of the circuit is determined by a linear fit on the linear range, giving $A = -6.19 \pm 0.131$. with error originatig only from the linear regression.

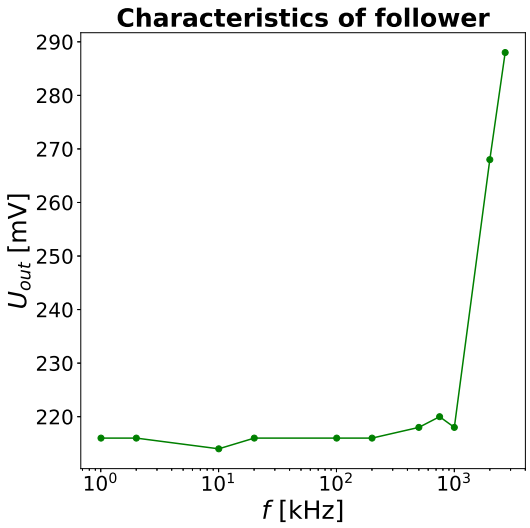


Figure 7: Output-voltage - frequency dependence of the follower with 210 mVpp sinusoidal input.

U_1 [mV]	U_2 [mV]	$U_{\text{out}}^{\text{expected}}$ [mV]	$U_{\text{out}}^{\text{measured}}$ [mV]
100 ± 50	100 ± 50	-294 ± 105	-266 ± 0.5
200 ± 50	100 ± 50	-441 ± 107	-324 ± 0.5
200 ± 50	200 ± 50	-588 ± 110	-542 ± 0.5
500 ± 50	500 ± 50	-1470 ± 138	-1414 ± 0.5
1000 ± 50	1000 ± 50	-2941 ± 208	-2926 ± 0.5

Table 1: Response of the summing amplifier for different inputs, with its calculated value.

3.3 M3 - Voltage follower

In this exercise we have built the voltage follower (voltage buffer) amplifier shown on Figure 3. We applied a 1 kHz 200 mVpp sinusoidal signal to its input. The response is shown on Figure 8. For 1 kHz input signal the circuit really behaves as a follower, since the amplitude and phase of the input and the output is similar.
Then we started increase the frequency of the input in order to find the cutoff (30% decrease in amplification). The output voltage by the frequency of the input is shown on Figure 7.
The behaviour of the circuit is not the expected, since the amplification started to increase. The achieveable frequency range provided by the function generator is not enough to reach the TL071 cutoff. The output behaviour at the maximum input frequency (2.7 MHz) is shown on Figure 9. The phase shift is clearly visible.

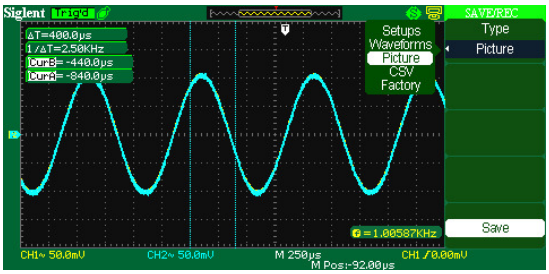


Figure 8: Input and output of the follower at 1 kHz. The amplitude and phase of the output is similar as the input.

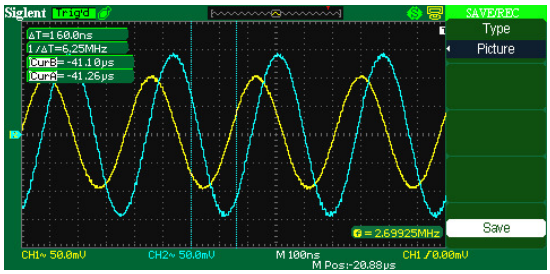


Figure 9: Input and output of the follower at

3.4 M4 - Differentiator

In this exercise we have built the differentiator circuit shown on Fig. 4. We have used $R_v = 100\,\mathrm{k\Omega}$, $C_p = 100\,\mathrm{pF}$ and $C = 10\,\mathrm{nF}$. We have applied 1 Vpp sinusoidal, triangular and square signals to its input and investigated its behaviour by (5). Are results are shown on figures 10, 12 and 14.

On the figures we can see, that for the sine signal there is a $\pi/2$ shift between the input and output, say the output is a (negative) cosine. The derivative of the triangle is a square, and the derivative of the square is a series of Dirac-deltas. Instead of deltas, we can see finite-amplitude pulses.

Now we removed C_p from the series. The results are shown on figures 11, 13 and 15. We can see, that the circuit gets more unstable against noise, since C_p acts as a low-pass filter.

3.5 M5 - Integrator

In this exercise, we have built the integrator circuit from Figure 5. Here $R_p = 10\,\mathrm{M\Omega}$, $C = 10\,\mathrm{nF}$ and $R_s = 100\,\mathrm{k\Omega}$. We applied 1 kHz 100 mVpp sine, triangle and square drivings, and our results are shown on figures 16, 18 and 20.

The output behaviour is described by (6). Here we can see the $\pi/2$ phase shift for the sine (the integral of the sine is a negative cosine). For the triangle we se prabolas and for the square we see a triangle.

We also built the circuit without R_p, the results are shown on figures 17, 19 and 21. Here the signals get noisier, since the high frequency cutoff is not provided.

4 Conclusions, outlook

In this measurement we investigated the basic properties of operational amplifier systems. At first, we built an inverting amplifier using a TL071 op-amp. The expected gain was $A = -6.8$, and our measurement provided by $A = -6.2 \pm 1.2$, so its measured value is the same as the calculated between ranges of error.

We also investigated the summing amplifier, and proved, that it provides the sum of the inputs weighted by the ratio of the input resistances by the feedback.

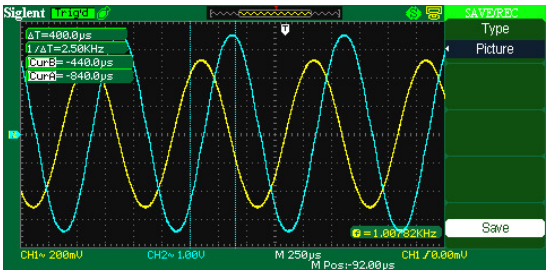


Figure 10: Output of differentiator for 1 kHz, 1 Vpp sinusoidal driving with C_p from Figure 4 connected.

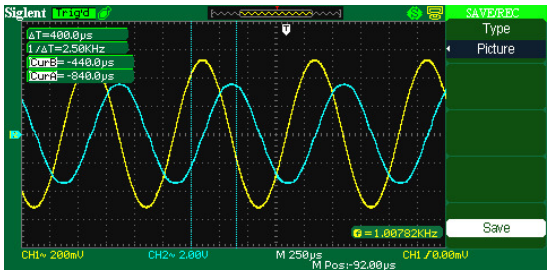


Figure 11: Output of differentiator for 1 kHz, 1 Vpp sinusoidal driving with C_p from Figure 4 disconnected.

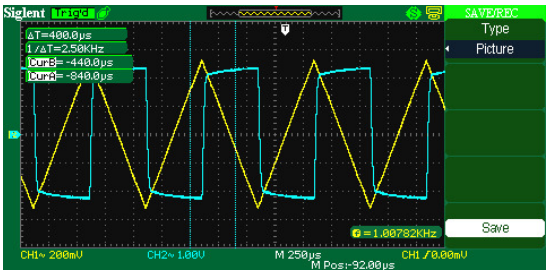


Figure 12: Output of differentiator for 1 kHz, 1 Vpp triangular driving with C_p from Figure 4 connected.

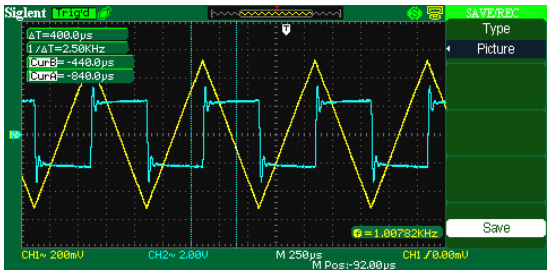


Figure 13: Output of differentiator for 1 kHz, 1 Vpp triangular driving with C_p from Figure 4 disconnected.

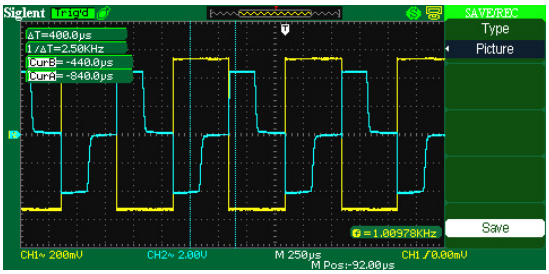


Figure 14: Output of differentiator for 1 kHz, 1 Vpp square driving with C_p from Figure 4 connected.

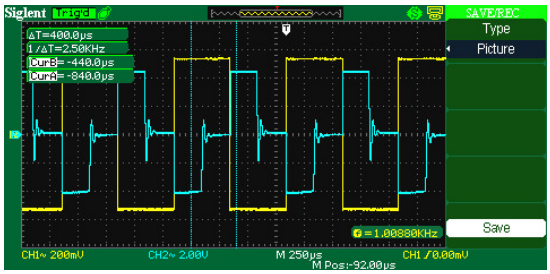


Figure 15: Output of differentiator for 1 kHz, 1 Vpp square driving with C_p from Figure 4 disconnected.

We proved that the voltage follower has $A = 1$ gain on 1 kHz, and it keeps the phase. As we started to increase the frequency, the amplitude started to increase instead of dropping, and we were unable to measure the cutoff of TL071.
We also investigated the differentiator and integrator circuits built from the TL071, and their expected behaviour is proven. We probed the circuits without the feedback capacitor (for the differentiator) and resistor (for the integrator) and we observed, that these devices work as low-passing filters.

References

[1] V. Farkas, M2 - Basic connections for operational amplifiers, measurement description, `https://oktatok.reak.bme.hu/farkas/wp-content/uploads/sites/`

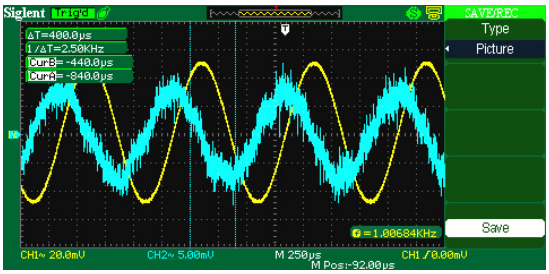


Figure 16: Output of integrator for 1 kHz, 100 mVpp sinusoidal driving with R_p from Figure 5 connected.

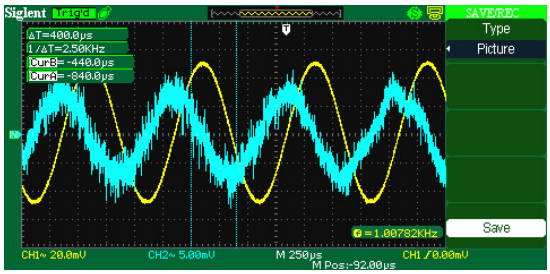


Figure 17: Output of integrator for 1 kHz, 100 mVpp sinusoidal driving with R_p from Figure 5 disconnected.

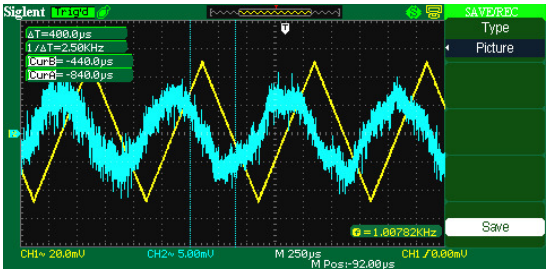


Figure 18: Output of integrator for 1 kHz, 100 mVpp triangular driving with R_p from Figure 5 connected.

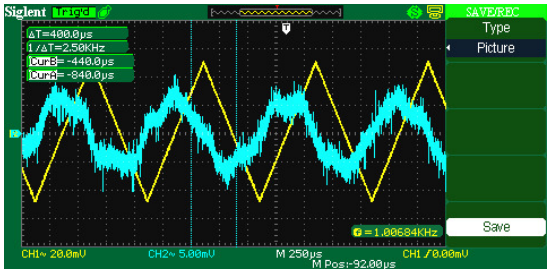


Figure 19: Output of integrator for 1 kHz, 100 mVpp triangular driving with R_p from Figure 5 disconnected.

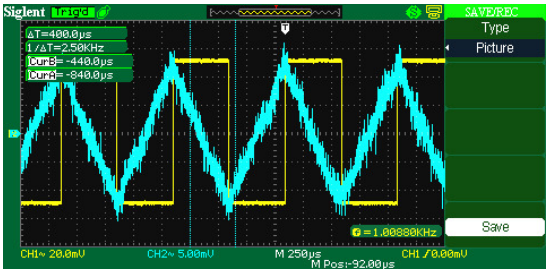


Figure 20: Output of integrator for 1 kHz, 100 mVpp square driving with R_p from Figure 5 connected.

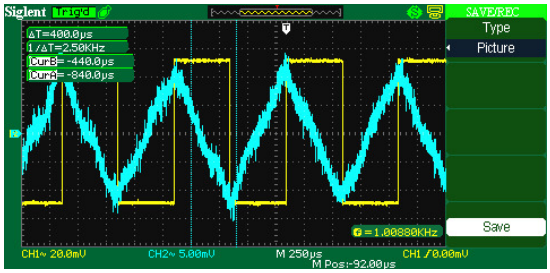


Figure 21: Output of integrator for 1 kHz, 100 mVpp square driving with R_p from Figure 5 disconnected.

14/2026/09/M2_OPAMP_20260827.pdf, (2026).

[2] TL071 operational amplifier data sheet, Texas Instruments, https://www.ti.com/lit/ds/symlink/tl071.pdf?ts=1790309131769&ref_url=https%253A%252F%252Fwww.ti.com%252Fproduct%252FTL071, (2026).

[3] Hameg HM 8040 triple power supply data sheet, Hameg Instruments, https://www.sm5cbw.se/hameg/hm80/hm8040-0m.pdf, (2026).

[4] Hameg HM 8011 digital multimeter data sheet, Hameg Instruments, https://www.sm5cbw.se/hameg/hm80/hm8012d.pdf, (2026).

[5] Hameg HM 8030 function generator data sheet, Hameg Instruments, https://physique.ens-lyon.fr/sites/default/files/materiel-physique-notices/N027-009.pdf, (2026).

[6] Siglent SDS 1072 CML oscilloscope data sheet, Hameg Instruments, https://www.wavecom.com.au/04_product_documents/1024/Brochure_SDS1072CML+_Ver_2.30.pdf?srsltid=AU7gw4WpjouW9-kxAKUSs7NsmQNXHRbCT_RvujsDPWRNXQojoCDG_jJV, (2026).